№ 11

$H_0: \ \xi \sim p_0(x) = 1\{(0;1)\}$

$H_1: \ \xi \sim p_1(x) = \dfrac{e^{1-x}}{e-1}\{(0;1)\}$

а) $\underline{n=1}$

$$l(x) = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq c \Rightarrow G: x \leq A \qquad P(x \leq A | H_0) = \alpha$$

$$\int_0^A 1dx = \alpha \Rightarrow A = \alpha \Rightarrow G: x \leq \alpha$$

$$W = P(x \leq A | H_1) = \int_0^A \frac{e^{1-x}}{e-1}dx = \frac{e}{e-1} \cdot \frac{e^{-x}}{(-1)}\Big|_0^\alpha = \frac{e}{e-1}(1 - e^{-\alpha})$$

$$\Rightarrow \alpha_1 = \alpha$$

$$\alpha_2 = 1 - W = \frac{\cancel{e} - 1 - \cancel{e} + e^{1-\alpha}}{e-1} = \frac{e^{1-\alpha} - 1}{e-1}$$

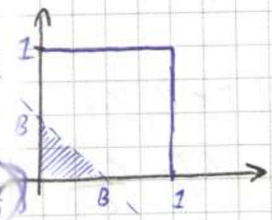
б) $\underline{n=2}$

$$\Rightarrow l = \frac{L_1}{L_0} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq c$$

$$\Rightarrow G: x_1 + x_2 \leq B = \sqrt{2\alpha}$$

$$P(x_1 + x_2 \leq B | H_0) = \alpha$$

$$\iint\limits_{\substack{x_1+x_2 \leq B \\ 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 1}} 1 \cdot 1 \cdot dx_1 dx_2 = \frac{B^2}{2} = \alpha \Rightarrow B = \sqrt{2\alpha}$$



$$; B \leq 1 \Rightarrow \sqrt{2\alpha} \leq 1 \Rightarrow \alpha \leq \frac{1}{2}$$

$$W = P(x_1 + x_2 \leq B | H_1) = \iint\limits_{x_1+x_2 \leq B} \frac{e^{1-x_1} e^{1-x_2}}{(e-1)^2} dx_1 dx_2 = \frac{e^2}{(e-1)^2} \int_0^B dx_1 \int_0^{B-x_1} e^{-x_1} e^{-x_2} dx_2 =$$

$$= \left(\frac{e}{e-1}\right)^2 \int_0^B e^{-x_1}\left(1 - e^{-B+x_1}\right) dx_1 = \left(\frac{e}{e-1}\right)^2 \left(1 - e^{-B} - Be^{-B}\right)$$

$$\alpha_1 = \alpha$$

$$\alpha_2 = 1 - W = 1 - \left(\frac{e}{e-1}\right)^2 \left(1 - e^{-B} - Be^{-B}\right)$$

c) асимптотич. критерий по выборке объема n:

$$l(\vec{x}_n) = \frac{L_1(\vec{x}_n)}{L_0(\vec{x}_n)} = \frac{\prod_{i=1}^{n} p_1(x_i)}{\prod_{i=1}^{n} p_0(x_i)} = \prod_{i=1}^{n}\left(\frac{p_1(x_i)}{p_0(x_i)}\right) \geq C$$

$$\ln l = \sum_{i=1}^{n} \underbrace{\ln\left(\frac{p_1(x_i)}{p_0(x_i)}\right)}_{=q_i} \geq \ln C$$

$$\sum_{i=1}^{n} \ln p_1(x_i) - \sum_{i=1}^{n} \ln p_0(x_i) = \sum_{i=1}^{n} \ln \frac{e^{1-x}}{e-1} - 0 \geq \ln C$$

$$\sum_{i=1}^{n}(1-x_i) - n\ln(e-1) \geq \ln C \Rightarrow \boxed{G: \sum_{i=1}^{n} x_i \leq A}$$

$$\bar{x} \leq B \qquad P(\bar{x} \leq B \mid H_0) = \alpha$$

Простейшая ЦПТ

$$\boxed{\begin{array}{l} M\xi = \frac{1}{2} \\ D\xi = \frac{1}{12} \end{array}}$$

$$\frac{\bar{\xi} - M\xi}{\sqrt{D\xi}}\sqrt{n} \rightsquigarrow N(0,1)$$

$$P(\bar{x} \leq B \mid H_0) = P\Bigg(\frac{\bar{x} - \frac{1}{2}}{\sqrt{\frac{1}{12}}}\sqrt{n} \leq \underbrace{\frac{B - \frac{1}{2}}{\sqrt{\frac{1}{12}}}\sqrt{n}}_{u_\alpha}\Bigg)$$

$$\Rightarrow B = \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$\Rightarrow G: \bar{x} \leq \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$W = P(\bar{x} \leq B \mid H_1)$$

$$H_1: M\xi = \frac{e-2}{e-1} \qquad M\xi^2 = \frac{2e-5}{e-1} \Rightarrow D\xi = M\xi^2 - (M\xi)^2 = \frac{e^2-3e+1}{(e-1)^2}$$

$\Rightarrow$ Снова ЦПТ:

$$W = P\Bigg(\frac{\bar{x} - M\xi}{\sqrt{D\xi}}\sqrt{n} \leq \underbrace{\frac{B - M\xi}{\sqrt{D\xi}}\sqrt{n}}_{\gamma}\Bigg) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

≈ 0.7804

$$\gamma = \frac{\frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}} - 0.418}{\sqrt{D\xi}}\sqrt{n} \xrightarrow[n \to +\infty]{} +\infty \Rightarrow W \xrightarrow[n \to \infty]{} 1$$

$$\alpha_1 = \alpha$$

$$\alpha_2 = 1 - W$$

d) Выборка объема n, $x_{min} < C$

$$F(x) = \int_{-\infty}^{x} p_0\,dt \qquad \Phi(x) = \int_{-\infty}^{x} p_1\,dt$$

$$\alpha_1 \le \alpha$$

$$\alpha_2 \to \min$$

$$\alpha_1 = P(x_{min} < C \mid H_0) = 1 - (1 - F(c))^n \le \alpha \Rightarrow (1 - F(c))^n \le \alpha - 1$$

$$1 - F(c) \ge \sqrt[n]{1-\alpha} \Rightarrow F(c) \le 1 - \sqrt[n]{1-\alpha}$$

$$\alpha_2 = P(x_{min} \ge C \mid H_1) = 1 - P(x_{min} < C \mid H_1) = 1 - 1 + (1 - \Phi(C))^n =$$
$$= (1 - \Phi(c))^n \to \min$$

$$\frac{\partial \alpha_2}{\partial C} = n\,(1-\Phi(c))^{n-1}(-p_1(c)) = \frac{-ne^{1-c}}{e-1}(1-\Phi(c))$$

$$\Phi(c) = \int_0^c p_1(x)dx = \frac{e}{e-1}\int_0^c e^{-x}dx = \frac{e(1-e^{-c})}{e-1}$$

$$1 - \Phi(c) = \frac{e - 1 + e^{1-c} - e}{e-1} = \frac{e^{1-c}-1}{e-1} \Rightarrow$$

$$\Rightarrow \frac{\partial \alpha_2}{\partial c} = -\frac{ne^{1-c}}{e-1}\left(\frac{e^{1-c}-1}{e-1}\right)^{n-1} < 0 \quad \forall c \in (0,1)$$

$\Rightarrow$ минимум на $c \le 1 - \sqrt[n]{1-\alpha}$ при $C = 1 - \sqrt[n]{1-\alpha}$

$$\Rightarrow G: x_{min} < 1 - \sqrt[n]{1-\alpha}$$

$$\alpha_2 = (1 - \Phi(1 - \sqrt[n]{1-\alpha}))^n = \left(\frac{1}{e-1}\left(e^{1-1+\sqrt[n]{1-\alpha}} - 1\right)\right)^n = \left(\frac{e^{\sqrt[n]{1-\alpha}}-1}{e-1}\right)^n$$

$$e^{\sqrt[n]{1-\alpha}} = e^{1+\frac{1}{n}\ln(1-\alpha)+o(\frac{1}{n})} = e\left(1 + \frac{1}{n}\ln(1-\alpha) + o(\tfrac{1}{n})\right) \Big\} \Rightarrow$$

$$\Rightarrow \alpha_2 = \left(\frac{e\left(1 + \frac{1}{n}\ln(1-\alpha) + o(\frac{1}{n})\right) - 1}{e-1}\right)^n = \left(1 + \frac{e\ln(1-\alpha)}{(e-1)n} + o(\tfrac{1}{n})\right)^n \Rightarrow$$

$$\Rightarrow \lim_{n\to\infty} \alpha_2 = \lim_{n\to\infty}\left(1 + \frac{e\ln(1-\alpha)}{(e-1)n} + o(\tfrac{1}{n})\right)^n = \exp\left[\frac{e\ln(1-\alpha)}{e-1}\right] =$$
$$= (1-\alpha)^{\frac{e}{e-1}}$$

$$\Rightarrow \lim_{n\to\infty} W = \lim_{n\to\infty}(1 - \alpha_2) = 1 - (1-\alpha)^{\frac{e}{e-1}} \ne 1 \Rightarrow$$

Критерий не явл. состоятельным

$$\alpha_1 = 1 - (1 - F(1 - \sqrt[n]{1-\alpha}))^n = 1 - (\sqrt[n]{1-\alpha})^n = \alpha$$

$$\alpha_2 = \left(\frac{e^{\sqrt[n]{1-\alpha}}-1}{e-1}\right)^n$$

№T12

$\alpha = 0.2$

$H_0: \ \xi \sim \frac{8}{24}\{4\} + \frac{4}{24}\{3\} + \frac{6}{24}\{1, 2\}$

$H_1: \ \xi \sim \frac{1}{4}\{1, 2, 3, 4\}$

Пусть для $n=2$ тройка выпадет m_3 раз, четверка — m_4 раз. Тогда 1 или 2 — $n - m_3 - m_4$ раз $\Rightarrow$

$$\Rightarrow l = \frac{L_1}{L_0} = \frac{\left(\frac{1}{4}\right)^2}{\left(\frac{1}{3}\right)^{m_4}\left(\frac{1}{6}\right)^{m_3}\left(\frac{1}{4}\right)^{2-m_4-m_3}} = \frac{\left(\frac{1}{4}\right)^{m_3+m_4}}{\left(\frac{1}{3}\right)^{m_4}\left(\frac{1}{6}\right)^{m_3}} =$$

$$= \frac{\left(\frac{1}{2}\right)^{2m_3+2m_4}}{\left(\frac{1}{3}\right)^{m_4}\left(\frac{1}{3}\right)^{m_3}\left(\frac{1}{2}\right)^{m_3}} = \frac{\left(\frac{1}{2}\right)^{2m_4-m_3}}{\left(\frac{1}{3}\right)^{m_3+m_4}} = \frac{2^{m_3-2m_4} \cdot 3^{m_3+m_4}}{\cancel{8}}$$

N	1	2	3	4	5	6
(m_3, m_4)	$(0,0)$	$(0, 1)$	$(0, 2)$	$(1, 0)$	$(1, 1)$	$(2, 0)$
l	1	$\frac{3}{4}$	$\frac{9}{16}$	$\frac{2}{3}$	$\frac{9}{8}$	$\frac{4}{9}$
$H_0 \to P$	$1/4$	$1/3$	$1/9$	$\frac{1}{6}$	$1/9$	$1/36$
$H_1 \to P$	$1/4$	$1/4$	$1/16$	$1/4$	$1/8$	$1/16$
Исходы:	4	4	1	4	2	1

$P(l \geq c \mid H_0) \leq \alpha = 0.2$

(6) $p = \frac{1}{36} \leq 0.2$

(5) $p = \frac{1}{36} + \frac{1}{6} + \frac{1}{9} = \frac{11}{36} > 0.2$?!

(4) $p = \frac{1}{36} + \frac{1}{6} = \frac{7}{36} \leq 0.2$

$\Rightarrow G: \ m_3 \geq 1$

$\alpha_1 = \frac{7}{36}$; $W = P(l \geq c \mid H_1) = \frac{1}{4} + \frac{1}{16} = \frac{5}{16} \Rightarrow$

$\Rightarrow \alpha_2 = 1 - W = \frac{9}{16}$

№13

$\xi \sim N(\theta, a^2)$ $\qquad$ $\eta \sim N(4\ \theta b^2)$

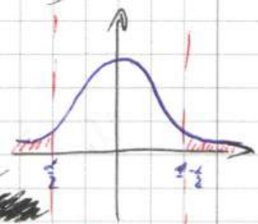
$\vec{x}_{139}$ $\qquad$ $\vec{y}_{100}$

$H_0:\ a^2 = b^2$

$H_1:\ \overline{H_0}$ $\qquad$ $\alpha = 0.05$

$$\frac{(n-1)S_x^2}{a^2} \sim \chi^2(n-1), \quad \frac{(m-1)S_y^2}{b^2} \sim \chi^2(m-1) \Rightarrow$$

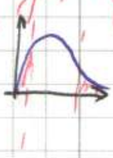
$$\Rightarrow \Delta = \frac{\frac{S_x^2}{a^2}}{\frac{S_y^2}{b^2}} \sim F(n-1, m-1)$$



$\hat{\Delta} = \frac{S_x^2}{S_y^2}$, если H_0 верна. $\Rightarrow G:\ \hat{\Delta} \le F_{\frac{\alpha}{2}}(n-1, m-1)$

$\hat{\Delta} > F_{1-\frac{\alpha}{2}}(n-1, m-1)$

P.S. Точно будет распред. Фишера:



$$\Rightarrow G: \left[\begin{array}{l} \hat{\Delta} \le 0.77 \\ \hat{\Delta} > 1.27 \end{array}\right.$$

Для длины $\hat{\Delta} = 0.86$

Для ширины $\hat{\Delta} = 0.83$ $\quad$ $\Rightarrow$ Нельзя отвергать H_0

Пусть $\theta = \frac{a^2}{b^2}$

$$W(\theta) = P\left(\frac{S_x^2}{S_y^2} \le 0.77 \Big| H_1\right) + P\left(\frac{S_x^2}{S_y^2} > 1.27 \Big| H_1\right) = P\left(\frac{S_x^2}{\theta S_y^2} \le \frac{0.77}{\theta}\right) + P\left(\frac{S_x^2}{\theta S_y^2} > \frac{1.27}{\theta}\right) =$$

$$= \int_0^{\frac{0.77}{\theta}} q(x)\,dx + \int_{\frac{1.27}{\theta}}^{+\infty} q(x)\,dx.$$ График надо.

№T14

$\xi \sim N(a, 2)$

$\eta \sim N(b, 1)$ $\quad$ | независимы.

$\vec{x}_n = \{-1.11;\ -0.1;\ 2.42\}$

$\vec{y}_m = \{-2.79;\ -7.91\}$

$$\boxed{\begin{array}{l} H_0:\ a = b \\ H_1:\ a > b \end{array}}$$

$\frac{\bar{x}-a}{\sigma_x}\sqrt{n} \sim N(0,1)$ $\qquad$ $\frac{\bar{y}-b}{\sigma_y}\sqrt{m} \sim N(0,1)$

$\Rightarrow \bar{x}-a \sim N(0; \frac{\sigma_x^2}{n})$ $\qquad$ $\bar{y}-b \sim N(0, \frac{\sigma_y^2}{m})$

$\Rightarrow \bar{x}-a-(\bar{y}-b) \sim N(0; \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}) = N(0; \frac{7}{6})$

$\Delta = \frac{\bar{x}-\bar{y}-(a-b)}{\sqrt{\frac{7}{6}}} \sim N(0,1)$

$H_0 \to \hat{\Delta} = \frac{\bar{x}-\bar{y}}{\sqrt{\frac{7}{6}}} \sim N(0;1)$ $\qquad$ G $\hat{\Delta} \geq u_{1-\alpha} = 1.645$

$\cancel{\bar{x} = -1{,}597 \quad \bar{y} = 2.6}$ $\Rightarrow$ $\hat{\Delta}\ 0.929 < 1.645 \Rightarrow$ Нет основ. отвергать H_0

p-value $= P(\Delta \geq \hat{\Delta} | H_0) = 1 - \Phi(0.929) = 0.177 > 0.05 \Rightarrow$ Нет оснований отвергать H_0

$W = P(\hat{\Delta} \geq 1{,}645 | H_1) = P(\frac{\bar{x}-\bar{y}}{\sqrt{\frac{7}{6}}} \geq 1.645 | H_1) =$

$= P(\frac{\bar{x}-\bar{y}-\theta}{\sqrt{\frac{7}{6}}} \geq 1.645 - \frac{\theta}{\sqrt{\frac{7}{6}}} | H_1) = 1 - \Phi(1.645 - \frac{\theta}{\sqrt{\frac{7}{6}}})$, $\theta = a-b$

$\Rightarrow W(\theta) = 1 - \int_{\infty}^{1.645 - \frac{\theta}{\sqrt{7/6}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$

Графики Аналогично Кадом